

Application of Information & Communication Technologies

Lecture-9

Recap of Lecture 8

- ◆ Secondary Storage
 - Partitioning and File Systems
 - Optical Discs and Drives
- ◆ Network Cloud Storage
- ◆ Other types of Storage Systems

Overview of Lecture 9

- ◆ Input Devices
- ◆ How humans give data and instructions to the computer

How humans give data and instructions to the computer



What are Input Devices?

- ◆ Devices we use to send data to the computer.
- ◆ Without input → computer cannot work.
- ◆ Examples: keyboard, mouse, scanner, camera, mic.

Keyboard Basics

- ◆ Most common input device.
- ◆ An input device used to enter characters at the location marked by the insertion point or cursor
- ◆ Layout: QWERTY.
- ◆ Keys: letters, numbers, function keys, control keys.

Keyboard Basics

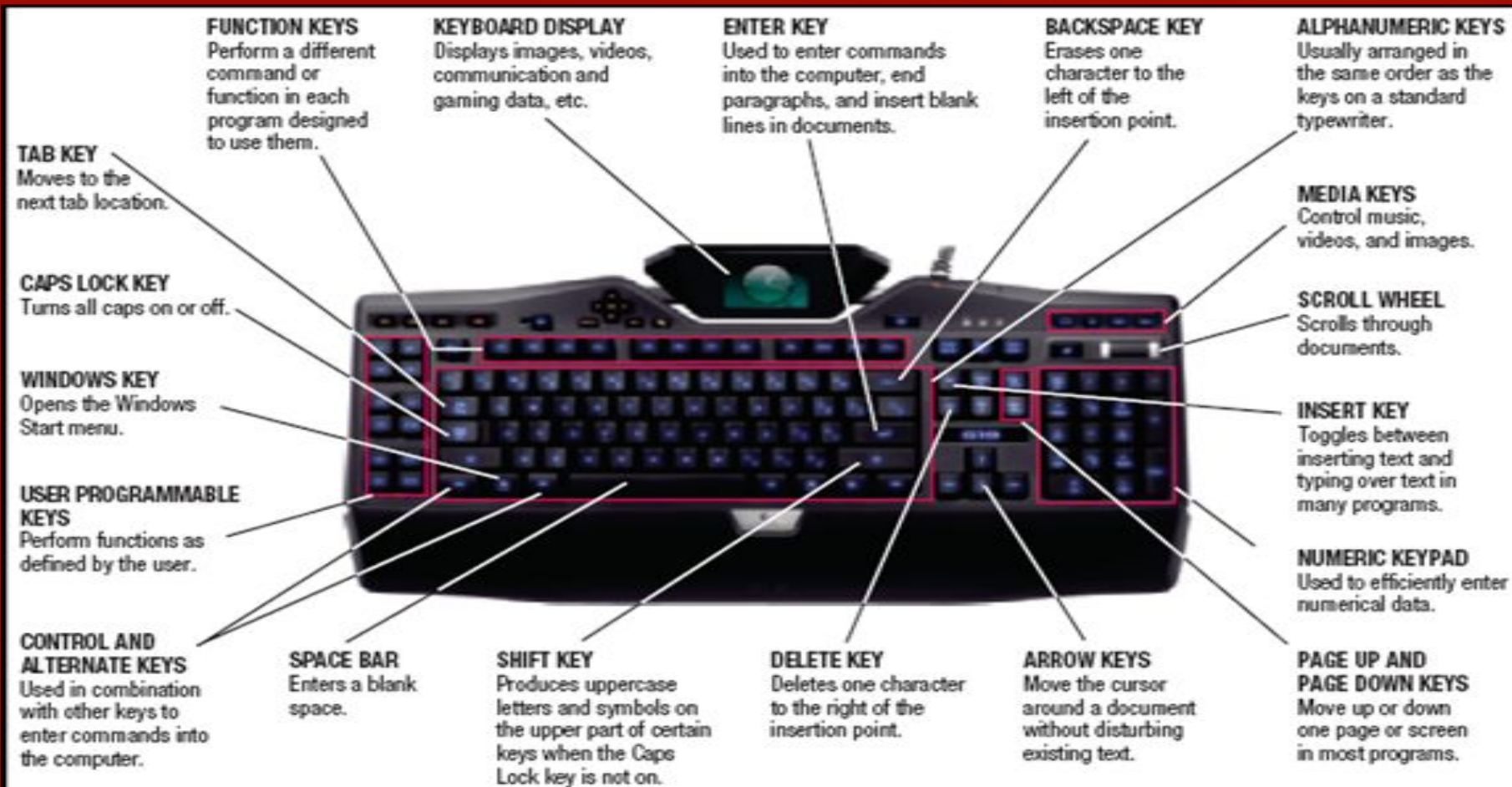


FIGURE 4-1

A typical desktop keyboard.

Types of Keyboards

- ◆ Standard keyboard – normal PC keyboard.
- ◆ Multimedia keyboard – extra keys (volume, media).
- ◆ Virtual keyboard – on screen, in mobiles.
- ◆ Wireless keyboard – no cable, uses Bluetooth.
- ◆ Usage
 - Typing text and numbers.
 - Entering commands.
 - Shortcuts (Ctrl+C = Copy, Ctrl+V = Paste).

Pointing Devices

- ◆ Device that moves the pointer/cursor on screen.
- ◆ Example:
 - mouse
 - Touch screen
 - Touchpad
 - Pen/stylus

Mouse

- ◆ Main pointing device.
- ◆ Actions: point, click, drag & drop.
- ◆ Types: mechanical, optical, wireless.
 - Older mechanical mice use a ball
 - Optical or laser mice track with light
 - 3D mice

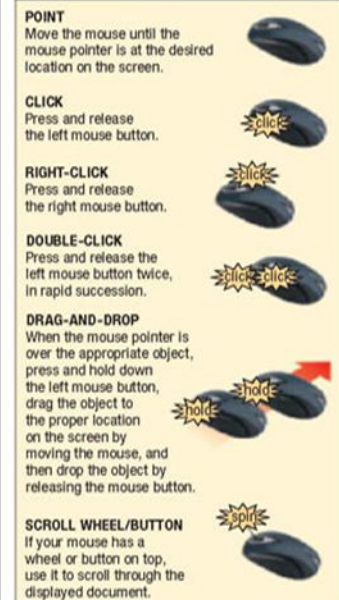


FIGURE 4-4
Common mouse operations.

Touchpad & Trackball

- ◆ **Touchpad** → flat area, used in laptops.
- ◆ **Trackball** → ball on top, move with fingers.



Stylus & Touchscreen

- ◆ Stylus → pen-like device for tablets.



Stylus & Touchscreen

- ◆ Touchscreen → finger as input (notebook, phones, ATMs).



Figure 5-8 This traveler checks in using an airport kiosk.



Scanner

- ◆ Scanner (optical scanner): Input device that captures an image of an object and transfers them to a computer in digital form
 - Can scan photos, documents, drawings, (flat objects)
 - Data is typically input as a single image
 - If optical character recognition (OCR) is used, text is input as individual text characters
- ◆ Types: flatbed, handheld, sheet-fed.



Capacity Comparison

- ◆ Optical resolution: Quality of scanned images
 - Measured in number of dots per inch (dpi)
 - Can often be specified when image is scanned
 - Can be changed when scanned image is edited
 - Varies with scanner used
 - Higher resolution = better quality but larger file size



Barcode Readers

- ◆ Barcode readers: Input devices that read barcodes
- ◆ Barcode: Machine-readable code that represents data as a set of bars
 - Common types
 - Universal Product Code (UPC)
 - ISBN
 - Code 39
 - Intelligent mail code
 - 2D (QR) – hold more data

Barcode & QR Code Readers

- ◆ Reads black & white bars.
 - supermarkets.
- ◆ Reads square codes (QR codes).
 - mobile payments, apps.



Biometric Input

- ◆ Uses body traits to identify people.
- ◆ Examples:
 - Fingerprint
 - Face geometry
 - Eye iris
 - Voice / Signature
- ◆ Biometric Readers:
 - Can be stand-alone (separate device) or built-in (e.g., in laptops, phones).
 - Allow access only to authorized users.
 - Common use → security, logins, payment verification.



Digital Cameras

◆ Still Cameras (Photos)

- Capture images (measured in megapixels).
- Use flash memory cards (SD cards).
- Built into mobile phones.
- Can also scan barcodes, mobile deposits.



◆ Video Cameras (Recording)

- Digital camcorders, PC webcams.
- Store videos on flash memory, DVD, or hard drive.
- Applications: surveillance, video calls, webinars, face recognition.



Webcam & Microphone

◆ Webcam

- Small camera for video calls.
- Built-in or external.



◆ Microphone

- Captures sound.
- Uses: recording, voice chat, voice commands.



Summary Table

Device	Example Use
Keyboard	Typing
Mouse	Point & click
Scanner	Paper → digital
Camera	Capture images
Mic	Voice input

Activity (Class)

- ◆  Look at your laptop/phone.
- ◆ Which input devices do you use every day?
- ◆ Can you work without a keyboard? Why/why not?

Quick Quiz

- ◆ Give two examples of pointing devices.
- ◆ Which device converts paper into digital form?
- ◆ Name one biometric input device.

Summary

- ◆ Input devices = bridge between humans & computers.

Suggested Reading

- ◆ Ch-04, The System Unit: Processing and Memory ,
“Understanding Computers: Today and Tomorrow,
Comprehensive”, 15th Edition by Deborah Morley
& Charles S. Parker
- ◆ Ch-05, Discovering Computers Fundamentals-
Your Interactive Guide -- Gary B Shelly; Misty E
Vermaat; Jeffrey J Q